

| UC-ID      | Use Case                      | Requirement | Priority |
|------------|-------------------------------|-------------|----------|
| UC-04      | Tạo công việc                 | F-03        | High     |
| UC-05      | Chỉnh sửa công việc           | F-03        | High     |
| UC-06      | Xóa công việc                 | F-03        | High     |
| UC-07      | Phân công thành viên          | F-03        | Medium   |
| UC-08      | Quản lý Label                 | F-03        | Medium   |
| UC-09      | Thiết lập Due Date            | F-03        | Medium   |
| UC-10      | Tạo List                      | F-02        | High     |
| UC-11      | Chỉnh sửa List                | F-02        | High     |
| UC-12      | Thay đổi danh sách công việc  | F-03        | High     |
| UC-13      | Xóa List                      | F-02        | High     |
|            |                               |             |          |
| *Chú thích |                               |             |          |
| F-03       | Quản lý công việc (Card/Task) |             |          |
| F-02       | Quản lý bảng và danh sách     |             |          |

| UC-ID | Service                    | Method                                           |
|-------|----------------------------|--------------------------------------------------|
| UC-04 | CardService                | createCard                                       |
| UC-05 | CardService                | updateCard                                       |
| UC-06 | CardService                | deleteCard                                       |
| UC-07 | CardService                | assignMember                                     |
| UC-08 | CardService + LabelService | createAndAttachLabel / attachLabel / detachLabel |
| UC-09 | CardService                | updateDueDate                                    |
| UC-10 | ListService                | createList                                       |
| UC-11 | ListService                | updateList                                       |
| UC-12 | CardService                | moveCard                                         |
| UC-13 | ListService                | deleteList                                       |

| UT-ID  | UC    | Business rule          | Service Method | Scenario             | Expected                  |
|--------|-------|------------------------|----------------|----------------------|---------------------------|
| UT-001 | UC-04 | Title required         | createCard     | title=""             | Throw Validation Error    |
| UT-002 | UC-04 | Title not whitespace   | createCard     | title=" "            | Throw Validation Error    |
| UT-003 | UC-04 | List exists            | createCard     | invalid listId       | Throw NotFound            |
| UT-004 | UC-04 | Create success         | createCard     | valid data           | Card created              |
| UT-005 | UC-04 | Activity logging       | createCard     | create success       | Activity created          |
| UT-006 | UC-05 | Card exists            | updateCard     | invalid cardId       | Throw NotFound            |
| UT-007 | UC-05 | Title required         | updateCard     | title=""             | Throw Validation Error    |
| UT-008 | UC-05 | Title not whitespace   | updateCard     | title=" "            | Throw Validation Error    |
| UT-009 | UC-05 | Empty patch handling   | updateCard     | body={}              | Return existing card      |
| UT-010 | UC-05 | Update title           | updateCard     | valid title          | Title updated             |
| UT-011 | UC-05 | Update description     | updateCard     | valid description    | Description updated       |
| UT-012 | UC-05 | Update multiple fields | updateCard     | title + description  | Fields updated            |
| UT-013 | UC-05 | Activity logging       | updateCard     | update success       | Activity created          |
| UT-014 | UC-06 | Card exists            | deleteCard     | invalid cardId       | Throw NotFound            |
| UT-015 | UC-06 | Soft delete            | deleteCard     | valid card           | deletedAt populated       |
| UT-016 | UC-06 | Activity logging       | deleteCard     | delete success       | Archived activity created |
| UT-017 | UC-12 | Target list required   | moveCard       | targetListId missing | Throw Validation Error    |
| UT-018 | UC-12 | newIndex valid         | moveCard       | newIndex < 0         | Throw Validation Error    |
| UT-019 | UC-12 | Card exists            | moveCard       | invalid cardId       | Throw NotFound            |
| UT-020 | UC-12 | Target list exists     | moveCard       | invalid targetListId | Throw NotFound            |

| UT-ID  | UC    | Business rule           | Service Method       | Scenario                    | Expected                   |
|--------|-------|-------------------------|----------------------|-----------------------------|----------------------------|
| UT-021 | UC-12 | Same board validation   | moveCard             | target list different board | Throw Validation Error     |
| UT-022 | UC-12 | Move same list          | moveCard             | same list valid move        | Order updated              |
| UT-023 | UC-12 | Reorder same list       | moveCard             | reorder position            | Index updated              |
| UT-024 | UC-12 | Move another list       | moveCard             | different list              | List updated               |
| UT-025 | UC-12 | Activity logging        | moveCard             | move success                | Activity created           |
| UT-026 | UC-09 | Due date required       | patchDueDate         | dueDate missing             | Throw Validation Error     |
| UT-027 | UC-09 | Valid ISO date          | patchDueDate         | invalid date string         | Throw Validation Error     |
| UT-028 | UC-09 | Card exists             | patchDueDate         | invalid cardId              | Throw NotFound             |
| UT-029 | UC-09 | Add due date            | patchDueDate         | card without due date       | Due date added             |
| UT-030 | UC-09 | Update due date         | patchDueDate         | existing due date           | Due date updated           |
| UT-031 | UC-09 | Remove due date         | deleteDueDate        | card has due date           | Due date removed           |
| UT-032 | UC-09 | Empty due date handling | deleteDueDate        | card has no due date        | Return existing card       |
| UT-033 | UC-08 | Label name required     | createAndAttachLabel | name missing                | Throw Validation Error     |
| UT-034 | UC-08 | Card exists             | createAndAttachLabel | invalid cardId              | Throw NotFound             |
| UT-035 | UC-08 | Create label success    | createAndAttachLabel | valid data                  | Label created and attached |
| UT-036 | UC-08 | Activity logging        | createAndAttachLabel | create success              | Activity created           |
| UT-037 | UC-08 | LabelId required        | attachLabel          | labelId missing             | Throw Validation Error     |
| UT-038 | UC-08 | Label exists            | attachLabel          | invalid labelId             | Throw NotFound             |
| UT-039 | UC-08 | Prevent duplicate       | attachLabel          | label already attached      | Throw Conflict             |
| UT-040 | UC-08 | Attach success          | attachLabel          | valid label                 | Label attached             |

| UT-ID  | UC    | Business rule           | Service Method | Scenario                        | Expected               |
|--------|-------|-------------------------|----------------|---------------------------------|------------------------|
| UT-041 | UC-08 | Label assignment exists | detachLabel    | label not attached              | Throw NotFound         |
| UT-042 | UC-08 | Detach success          | detachLabel    | valid label                     | Label detached         |
| UT-043 | UC-07 | Member required         | addMember      | workspaceMemberPublicId missing | Throw Validation Error |
| UT-044 | UC-07 | Card exists             | addMember      | invalid cardId                  | Throw NotFound         |
| UT-045 | UC-07 | Prevent duplicate       | addMember      | member already assigned         | Throw Conflict         |
| UT-046 | UC-07 | Assign success          | addMember      | valid member                    | Member assigned        |
| UT-047 | UC-07 | Activity logging        | addMember      | assign success                  | Activity created       |
| UT-048 | UC-07 | Member required         | removeMember   | workspaceMemberPublicId missing | Throw Validation Error |
| UT-049 | UC-07 | Assignment exists       | removeMember   | member not assigned             | Throw NotFound         |
| UT-050 | UC-07 | Remove success          | removeMember   | valid member                    | Member removed         |
| UT-051 | UC-10 | Name required           | createList     | body={}                         | Throw Validation Error |
| UT-052 | UC-10 | Name must be string     | createList     | name=123                        | Throw Validation Error |
| UT-053 | UC-10 | Name not whitespace     | createList     | name=" "                        | Throw Validation Error |
| UT-054 | UC-10 | Board exists            | createList     | invalid boardId                 | Throw NotFound         |
| UT-055 | UC-10 | Trim name               | createList     | name=" Todo "                   | Saved as "Todo"        |
| UT-056 | UC-10 | Create success          | createList     | valid data                      | List created           |
| UT-057 | UC-11 | Name required           | updateList     | body={}                         | Throw Validation Error |
| UT-058 | UC-11 | Name must be string     | updateList     | name=123                        | Throw Validation Error |
| UT-059 | UC-11 | Name not whitespace     | updateList     | name=" "                        | Throw Validation Error |
| UT-060 | UC-11 | List exists             | updateList     | invalid listId                  | Throw NotFound         |

| UT-ID  | UC    | Business rule       | Service Method | Scenario         | Expected               |
|--------|-------|---------------------|----------------|------------------|------------------------|
| UT-061 | UC-11 | Trim name           | updateList     | name=" Doing "   | Saved as "Doing"       |
| UT-062 | UC-11 | Update success      | updateList     | valid data       | List updated           |
| UT-063 | UC-13 | List exists         | deleteList     | invalid listId   | Throw NotFound         |
| UT-064 | UC-13 | Soft delete         | deleteList     | valid listId     | List soft deleted      |
| UT-065 | UC-08 | Name required       | createLabel    | body={}          | Throw Validation Error |
| UT-066 | UC-08 | Name must be string | createLabel    | name=123         | Throw Validation Error |
| UT-067 | UC-08 | Name not whitespace | createLabel    | name=" "         | Throw Validation Error |
| UT-068 | UC-08 | Board exists        | createLabel    | invalid boardId  | Throw NotFound         |
| UT-069 | UC-08 | Trim name           | createLabel    | name=" Bug "     | Saved as "Bug"         |
| UT-070 | UC-08 | Create success      | createLabel    | valid data       | Label created          |
| UT-071 | UC-08 | Name not empty      | updateLabel    | name=""          | Throw Validation Error |
| UT-072 | UC-08 | Name not whitespace | updateLabel    | name=" "         | Throw Validation Error |
| UT-073 | UC-08 | Board exists        | updateLabel    | invalid boardId  | Throw NotFound         |
| UT-074 | UC-08 | Label exists        | updateLabel    | invalid labelId  | Throw NotFound         |
| UT-075 | UC-08 | Trim name           | updateLabel    | name=" Feature " | Saved as "Feature"     |
| UT-076 | UC-08 | Update name         | updateLabel    | valid name       | Label updated          |
| UT-077 | UC-08 | Update colour       | updateLabel    | valid colourCode | Colour updated         |
| UT-078 | UC-08 | Board exists        | deleteLabel    | invalid boardId  | Throw NotFound         |
| UT-079 | UC-08 | Label exists        | deleteLabel    | invalid labelId  | Throw NotFound         |
| UT-080 | UC-08 | Soft delete         | deleteLabel    | valid label      | Label soft deleted     |

| UT-ID          | UC        | Business rule | Service Method | Scenario | Expected |
|----------------|-----------|---------------|----------------|----------|----------|
|                |           |               |                |          |          |
|                |           |               |                |          |          |
| <b>Service</b> | <b>UT</b> |               |                |          |          |
| CardService    | 50        |               |                |          |          |
| ListService    | 14        |               |                |          |          |
| LabelService   | 16        |               |                |          |          |
| <b>Total</b>   | <b>80</b> |               |                |          |          |

| IT-ID  | UC    | API                                     | Scenario                        | Expected Status | Expected Result            |
|--------|-------|-----------------------------------------|---------------------------------|-----------------|----------------------------|
| IT-001 | UC-04 | POST /lists/:listId/cards               | Valid request                   | 201             | Card created               |
| IT-002 | UC-04 | POST /lists/:listId/cards               | Empty title                     | 400             | Validation error           |
| IT-003 | UC-04 | POST /lists/:listId/cards               | Whitespace title                | 400             | Validation error           |
| IT-004 | UC-04 | POST /lists/:listId/cards               | Invalid listId                  | 404             | List not found             |
| IT-005 | UC-05 | PATCH /cards/:cardId                    | Update title                    | 200             | Title updated              |
| IT-006 | UC-05 | PATCH /cards/:cardId                    | Update description              | 200             | Description updated        |
| IT-007 | UC-05 | PATCH /cards/:cardId                    | Empty title                     | 400             | Validation error           |
| IT-008 | UC-05 | PATCH /cards/:cardId                    | Invalid cardId                  | 404             | Card not found             |
| IT-009 | UC-06 | DELETE /cards/:cardId                   | Delete success                  | 204             | Card deleted               |
| IT-010 | UC-06 | DELETE /cards/:cardId                   | Invalid cardId                  | 404             | Card not found             |
| IT-011 | UC-06 | DELETE /cards/:cardId                   | Delete card twice               | 404             | Card not found             |
| IT-012 | UC-07 | POST /cards/:cardId/members             | Missing workspaceMemberPublicId | 400             | Validation error           |
| IT-013 | UC-07 | POST /cards/:cardId/members             | Assign member success           | 201             | Member assigned            |
| IT-014 | UC-07 | DELETE /cards/:cardId/members/:memberId | Remove member success           | 204             | Member removed             |
| IT-015 | UC-07 | DELETE /cards/:cardId/members/:memberId | Member not assigned             | 404             | Assignment not found       |
| IT-016 | UC-08 | POST /boards/:boardId/labels            | Create label success            | 201             | Label created              |
| IT-017 | UC-08 | POST /boards/:boardId/labels            | Invalid boardId                 | 404             | Board not found            |
| IT-018 | UC-08 | POST /boards/:boardId/labels            | Empty label name                | 400             | Validation error           |
| IT-019 | UC-08 | POST /cards/:cardId/labels              | Attach existing label           | 201             | Label attached             |
| IT-020 | UC-08 | POST /cards/:cardId/labels              | Create and attach new label     | 201             | Label created and attached |
| IT-021 | UC-08 | POST /cards/:cardId/labels              | Invalid labelId                 | 404             | Label not found            |
| IT-022 | UC-08 | DELETE /cards/:cardId/labels/:labelId   | Detach label                    | 204             | Label detached             |



| IT-ID  | UC    | API                                                 | Scenario                 | Expected Status | Expected Result    |
|--------|-------|-----------------------------------------------------|--------------------------|-----------------|--------------------|
| IT-023 | UC-09 | PATCH /cards/:cardId/due-date                       | Valid due date           | 200             | Due date updated   |
| IT-024 | UC-09 | PATCH /cards/:cardId/due-date                       | Invalid cardId           | 404             | Card not found     |
| IT-025 | UC-09 | DELETE /cards/:cardId/due-date                      | Remove due date          | 200             | Due date removed   |
| IT-026 | UC-10 | POST /workspaces/:workspaceId/boards/:boardId/lists | Create list success      | 201             | List created       |
| IT-027 | UC-10 | POST /workspaces/:workspaceId/boards/:boardId/lists | Empty name               | 400             | Validation error   |
| IT-028 | UC-10 | POST /workspaces/:workspaceId/boards/:boardId/lists | Name is not string       | 400             | Validation error   |
| IT-029 | UC-10 | POST /workspaces/:workspaceId/boards/:boardId/lists | Invalid boardId          | 404             | Board not found    |
| IT-030 | UC-11 | PATCH /lists/:listId                                | Update title             | 200             | List updated       |
| IT-031 | UC-11 | PATCH /lists/:listId                                | Empty name               | 400             | Validation error   |
| IT-032 | UC-11 | PATCH /lists/:listId                                | Name is not string       | 400             | Validation error   |
| IT-033 | UC-11 | PATCH /lists/:listId                                | Invalid listId           | 404             | List not found     |
| IT-034 | UC-12 | PATCH /cards/:cardId/move                           | Invalid destination list | 404             | List not found     |
| IT-035 | UC-12 | PATCH /cards/:cardId/move                           | Invalid cardId           | 404             | Card not found     |
| IT-036 | UC-12 | PATCH /cards/:cardId/move                           | Missing targetListId     | 400             | Validation error   |
| IT-037 | UC-12 | PATCH /cards/:cardId/move                           | newIndex < 0             | 400             | Validation error   |
| IT-038 | UC-12 | PATCH /cards/:cardId/move                           | Move card another list   | 200             | Card moved         |
| IT-039 | UC-12 | PATCH /cards/:cardId/move                           | Reorder same list        | 200             | Card order updated |
| IT-040 | UC-13 | DELETE /lists/:listId                               | Delete list success      | 204             | List deleted       |
| IT-041 | UC-13 | DELETE /lists/:listId                               | Invalid listId           | 404             | List not found     |
| IT-042 | UC-13 | DELETE /lists/:listId                               | Delete list twice        | 404             | List not found     |
|        |       |                                                     |                          |                 |                    |
|        |       |                                                     |                          |                 |                    |
|        |       | UC                                                  | Số IT                    |                 |                    |

| IT-ID | UC | API                     | Scenario | Expected Status | Expected Result |
|-------|----|-------------------------|----------|-----------------|-----------------|
|       |    | UC-04 Create Card       | 4        |                 |                 |
|       |    | UC-05 Update Card       | 4        |                 |                 |
|       |    | UC-06 Delete Card       | 3        |                 |                 |
|       |    | UC-07 Member Assignment | 4        |                 |                 |
|       |    | UC-08 Label Management  | 7        |                 |                 |
|       |    | UC-09 Due Date          | 3        |                 |                 |
|       |    | UC-10 Create List       | 4        |                 |                 |
|       |    | UC-11 Update List       | 4        |                 |                 |
|       |    | UC-12 Move Card         | 6        |                 |                 |
|       |    | UC-13 Delete List       | 3        |                 |                 |
|       |    | Tổng                    | 42       |                 |                 |

| Req-ID | UC    | Use case                            | Service Method         | API                                                 | Functional Test | Unit Test | Integration Test |
|--------|-------|-------------------------------------|------------------------|-----------------------------------------------------|-----------------|-----------|------------------|
| F-03   | UC-04 | Tạo công việc                       | createCard()           | POST /lists/:listId/cards                           | TC-BS-01        | UT-004    | IT-001           |
| F-03   | UC-04 | Tiêu đề không được rỗng             | createCard()           | POST /lists/:listId/cards                           | TC-BS-02        | UT-001    | IT-002           |
| F-03   | UC-04 | Tiêu đề không chỉ chứa khoảng trắng | createCard()           | POST /lists/:listId/cards                           | TC-BS-04        | UT-002    | IT-003           |
| F-03   | UC-04 | List phải tồn tại                   | createCard()           | POST /lists/:listId/cards                           | -               | UT-003    | IT-004           |
| F-03   | UC-05 | Chỉnh sửa tiêu đề công việc         | updateCard()           | PATCH /cards/:cardId                                | TC-BS-05        | UT-010    | IT-005           |
| F-03   | UC-05 | Chỉnh sửa mô tả công việc           | updateCard()           | PATCH /cards/:cardId                                | TC-BS-07        | UT-011    | IT-006           |
| F-03   | UC-05 | Không cho phép title rỗng           | updateCard()           | PATCH /cards/:cardId                                | TC-BS-06        | UT-007    | IT-007           |
| F-03   | UC-05 | Card phải tồn tại                   | updateCard()           | PATCH /cards/:cardId                                | -               | UT-006    | IT-008           |
| F-03   | UC-06 | Xóa công việc thành công            | deleteCard()           | DELETE /cards/:cardId                               | TC-BS-09        | UT-015    | IT-009           |
| F-03   | UC-06 | Card phải tồn tại                   | deleteCard()           | DELETE /cards/:cardId                               | TC-BS-11        | UT-014    | IT-010           |
| F-03   | UC-06 | Không thể xóa card đã bị xóa        | deleteCard()           | DELETE /cards/:cardId                               | -               | -         | IT-011           |
| F-03   | UC-07 | workspaceMemberPublicId bắt buộc    | addMember()            | POST /cards/:cardId/members                         | -               | UT-043    | IT-012           |
| F-03   | UC-07 | Gán thành viên thành công           | addMember()            | POST /cards/:cardId/members                         | TC-BS-12        | UT-046    | IT-013           |
| F-03   | UC-07 | Gỡ thành viên thành công            | removeMember()         | DELETE /cards/:cardId/members/:memberId             | TC-BS-15        | UT-050    | IT-014           |
| F-03   | UC-07 | Thành viên phải được gán trước      | removeMember()         | DELETE /cards/:cardId/members/:memberId             | -               | UT-049    | IT-015           |
| F-03   | UC-08 | Tạo Label thành công                | createLabel()          | POST /boards/:boardId/labels                        | TC-BS-16        | UT-070    | IT-016           |
| F-03   | UC-08 | Board phải tồn tại                  | createLabel()          | POST /boards/:boardId/labels                        | -               | UT-068    | IT-017           |
| F-03   | UC-08 | Label name bắt buộc                 | createLabel()          | POST /boards/:boardId/labels                        | -               | UT-065    | IT-018           |
| F-03   | UC-08 | Gắn Label có sẵn                    | attachLabel()          | POST /cards/:cardId/labels                          | TC-BS-18        | UT-040    | IT-019           |
| F-03   | UC-08 | Tạo và gắn Label mới                | createAndAttachLabel() | POST /cards/:cardId/labels                          | TC-BS-16        | UT-035    | IT-020           |
| F-03   | UC-08 | Label phải tồn tại                  | attachLabel()          | POST /cards/:cardId/labels                          | -               | UT-038    | IT-021           |
| F-03   | UC-08 | Hủy Label khỏi Card                 | detachLabel()          | DELETE /cards/:cardId/labels/:labelId               | TC-BS-19        | UT-042    | IT-022           |
| F-03   | UC-09 | Thiết lập Due Date                  | patchDueDate()         | PATCH /cards/:cardId/due-date                       | TC-BS-21        | UT-029    | IT-023           |
| F-03   | UC-09 | Card phải tồn tại                   | patchDueDate()         | PATCH /cards/:cardId/due-date                       | -               | UT-028    | IT-024           |
| F-03   | UC-09 | Xóa Due Date                        | deleteDueDate()        | DELETE /cards/:cardId/due-date                      | TC-BS-23        | UT-031    | IT-025           |
| F-02   | UC-10 | Tạo List thành công                 | createList()           | POST /workspaces/:workspaceId/boards/:boardId/lists | TC-BS-29        | UT-056    | IT-026           |
| F-02   | UC-10 | Tên List bắt buộc                   | createList()           | POST /workspaces/:workspaceId/boards/:boardId/lists | TC-BS-30        | UT-051    | IT-027           |
| F-02   | UC-10 | Name phải là string                 | createList()           | POST /workspaces/:workspaceId/boards/:boardId/lists | -               | UT-052    | IT-028           |
| F-02   | UC-10 | Board phải tồn tại                  | createList()           | POST /workspaces/:workspaceId/boards/:boardId/lists | -               | UT-054    | IT-029           |

| Req-ID | UC    | Use case                    | Service Method | API                       | Functional Test    | Unit Test | Integration Test |
|--------|-------|-----------------------------|----------------|---------------------------|--------------------|-----------|------------------|
| F-02   | UC-11 | Chỉnh sửa List thành công   | updateList()   | PATCH /lists/:listId      | TC-BS-32           | UT-062    | IT-030           |
| F-02   | UC-11 | Tên List không được rỗng    | updateList()   | PATCH /lists/:listId      | TC-BS-34           | UT-057    | IT-031           |
| F-02   | UC-11 | Name phải là string         | updateList()   | PATCH /lists/:listId      | -                  | UT-058    | IT-032           |
| F-02   | UC-11 | List phải tồn tại           | updateList()   | PATCH /lists/:listId      | -                  | UT-060    | IT-033           |
| F-03   | UC-12 | Target List phải tồn tại    | moveCard()     | PATCH /cards/:cardId/move | -                  | UT-020    | IT-034           |
| F-03   | UC-12 | Card phải tồn tại           | moveCard()     | PATCH /cards/:cardId/move | -                  | UT-019    | IT-035           |
| F-03   | UC-12 | targetListId bắt buộc       | moveCard()     | PATCH /cards/:cardId/move | -                  | UT-017    | IT-036           |
| F-03   | UC-12 | newIndex phải hợp lệ        | moveCard()     | PATCH /cards/:cardId/move | -                  | UT-018    | IT-037           |
| F-03   | UC-12 | Di chuyển sang List khác    | moveCard()     | PATCH /cards/:cardId/move | TC-BS-25           | UT-024    | IT-038           |
| F-03   | UC-12 | Sắp xếp lại trong cùng List | moveCard()     | PATCH /cards/:cardId/move | TC-BS-26           | UT-023    | IT-039           |
| F-02   | UC-13 | Xóa List thành công         | deleteList()   | DELETE /lists/:listId     | TC-BS-35, TC-BS-36 | UT-064    | IT-040           |
| F-02   | UC-13 | List phải tồn tại           | deleteList()   | DELETE /lists/:listId     | -                  | UT-063    | IT-041           |
| F-02   | UC-13 | Không thể xóa List đã xóa   | deleteList()   | DELETE /lists/:listId     | -                  | -         | IT-042           |